

USER MANUAL

Automated Construction Management System

“Nong Nai Chang Khon Foo”

Project Members

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1. Introduction

"Nong Nai Chang Khon Foo" (นายช่างขนฟู) is an Automated Construction Management System designed to eliminate reporting delays in construction project management. Traditional site reporting relies on manual data entry or textual summaries, creating a significant "Data Lag" between on-site operations and the project management office. This system solves that problem by allowing site foremen to submit daily progress reports directly through the LINE messaging application — instantly routing data through an AI-powered pipeline to a centralized database and notifying the Project Engineer via email.

The system integrates four core platforms: LINE Official Account (user interface), n8n workflow orchestrator (automation engine), Google Sheets (Master Plan database), and Gmail (executive reporting channel).

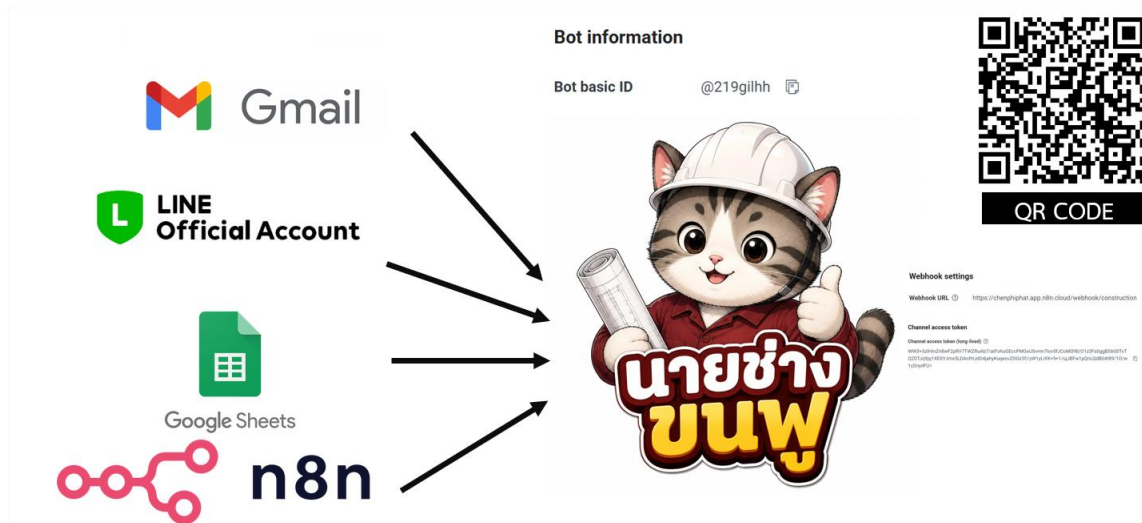


Figure 1: Overall system workflow — from site foreman (LINE) through n8n to Google Sheets and Gmail

1.1 Who Should Read This Manual

This manual is intended for the following users:

- **Site Foremen / โฟร์แมน** — Primary users who submit daily work progress reports via the LINE bot.
- **Project Engineers / เจ้าของงาน** — Supervisors who receive automated status reports via Gmail and monitor the Google Sheets Master Plan.
- **System Administrators** — Personnel responsible for managing the n8n workflow and LINE Official Account configuration.

1.2 System Requirements

Before using the system, ensure the following prerequisites are met:

- **LINE Application:** Installed on a smartphone (iOS or Android). Add the bot via QR code or Bot ID: **@219gilhh**
- **Internet Connection:** Required for all interactions — LINE, Google Sheets, and Gmail.
- **Gmail Account:** Project Engineer must have access to the designated Gmail inbox (chenphiphat.chen@gmail.com).
- **Google Sheets Access:** Required for monitoring the Master Plan database directly.

2. Getting Started

2.1 Adding the LINE Bot

Follow the steps below to connect to the Nong Nai Chang Khon Foo LINE Official Account:

1. Open the LINE application on your smartphone.
2. Tap the "Add Friend" icon (person with a + symbol) at the top of the screen.
3. Select "QR Code" to scan, or tap "Search" and enter the Bot ID: **@219gilhh**
4. Tap "Add" to add Nong Nai Chang Khon Foo as a friend.
5. Open the chat conversation with the bot to begin reporting.

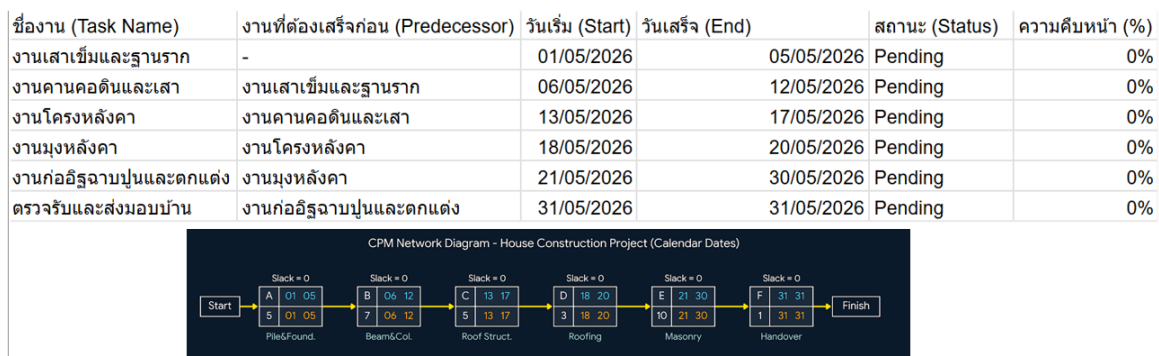


Figure 2: LINE application — adding the bot via QR code search using **@219gilhh**

2.2 Verifying the Connection

Once added, send the bot a greeting message such as "Hi" or "Hello". If the system is operational, you will receive an automated response prompting you to submit your work progress report. This confirms that your LINE account is successfully connected to the system.

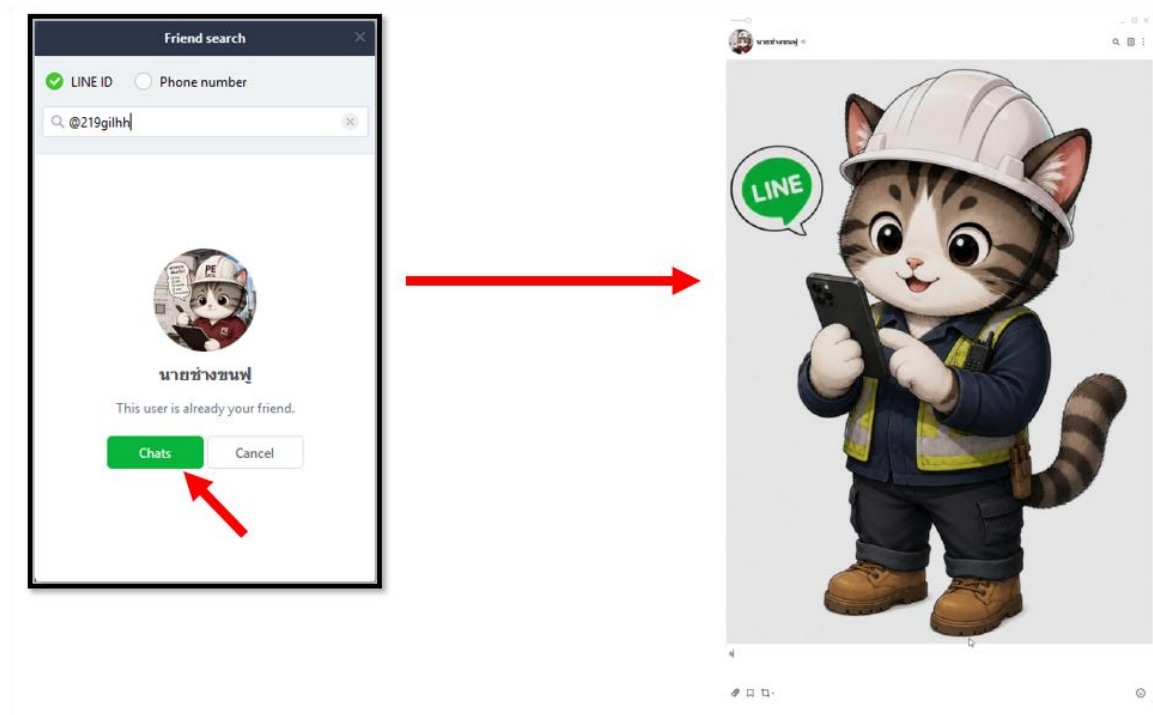


Figure 3: System response when a greeting message is sent — bot prompts user to submit a work report

3. Submitting a Work Progress Report

The primary function of this system is for site foremen to submit daily progress updates via the LINE bot. Reports are entered as natural language messages — there is no fixed form to fill in. The AI agent automatically extracts the required information from your message.

3.1 Required Information in Every Report

Every progress report message must contain the following three pieces of information:

1. Project Name (ชื่อโครงการ)

The name of the construction project being reported. Example: "บ้านตัวอย่าง ก" or "House Project A"

2. Date (วันที่)

The date on which the work was performed. Example: "1 May 2026" or "1 พ.ค. 2026"

3. Completion Percentage (ความคืบหน้า %)

The percentage of the reported task that has been completed. Example: "50%" or "งานเสร็จ 50 เปอร์เซ็นต์"

3.2 How to Format Your Message

You do not need to follow a strict template. Write your report naturally in Thai or English. The AI agent will understand your message and extract the required data automatically. However, make sure all three required fields (project name, date, and percentage) are clearly mentioned.

Example Report Messages (Thai):

อัครเดชงาน โครงการบ้านตัวอย่าง ก วันที่ 1 พ.ค. 2026 งานเสาเข็มและฐานราก เสร็จ 50%
โฟร์แมน ก รายงาน โครงการบ้านตัวอย่าง ก วันที่ 7 พ.ค. 2569 งานเสร็จ 100%

Example Report Messages (English):

Project: House A, Date: 1 May 2026, Foreman: A, Piling work 50% complete

3.3 What Happens After You Send a Report

Once your message is received, the system processes it automatically through the following steps:

6. **Your message is received** by the n8n Webhook and cleaned for processing.
7. **The AI Agent (Google Gemini) extracts** the project name, date, and completion % from your message.
8. **The AI compares** your reported progress against the CPM Master Plan schedule in Google Sheets.
9. **The Master Plan is updated** automatically with your new progress data.
10. **You receive a confirmation reply** on LINE acknowledging the update.
11. **The Project Engineer receives an email report** via Gmail containing a full analysis of the update.

4. Use Case Scenarios and Expected Results

This section demonstrates the four primary interaction scenarios between the site foreman and the system, along with the expected behavior and output of each.

4.1 Scenario 1: Greeting / System Test

When you send a simple greeting message (e.g., "Hi", "Hello", "สวัสดี") without any progress data, the system identifies this as a non-reporting interaction. It will not update any database and will respond with a friendly prompt asking you to submit a work report.

What to send:

"Hi" / "Hello" / "สวัสดี"

System response:

A LINE reply prompting you to submit a work report. No email is sent. No data is written.

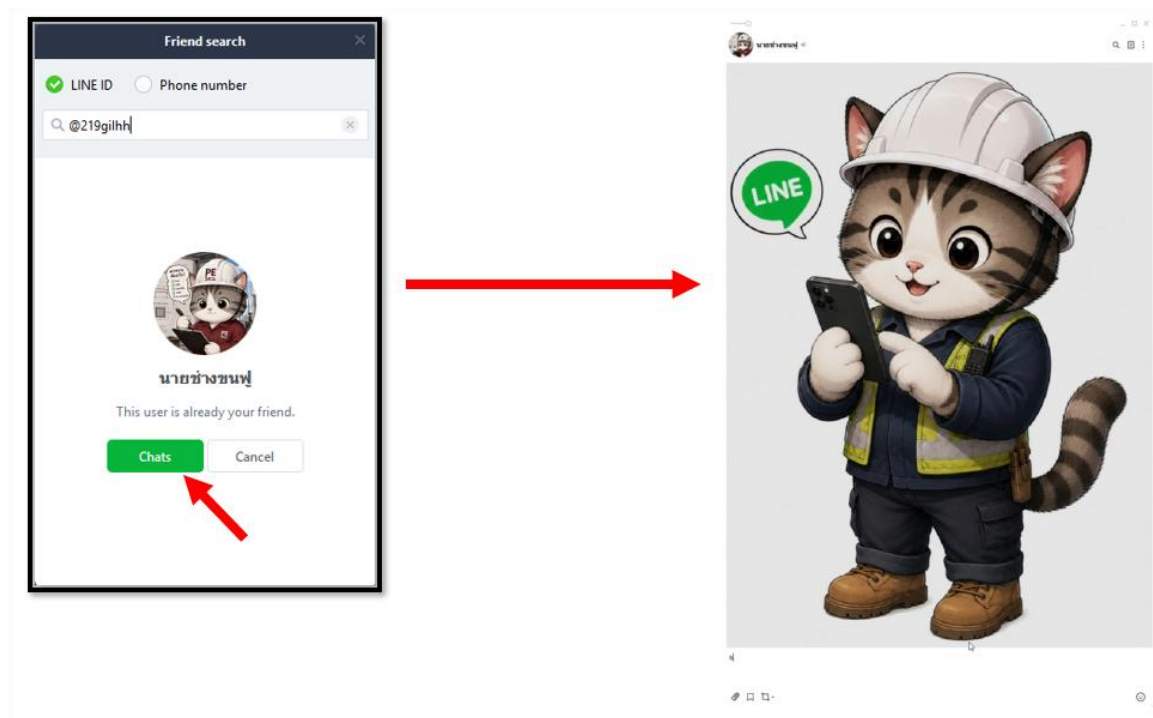


Figure 4: Scenario 1 — Bot response to a greeting message; no database update is triggered

4.2 Scenario 2: Normal Report — Ahead of Schedule

This is the standard daily reporting scenario. When a foreman submits a valid report with all required fields and the reported progress is equal to or ahead of the CPM baseline, the system processes the update and sends a positive confirmation.

Example message:

"อัปเดตงาน วันที่ 1 พ.ค. 2026 โครงการ บ้านตัวอย่าง ก โฟร์แมน ก งานเสาเข็มและฐานราก เสร็จ 50%"

What happens:

AI extracts data → checks CPM baseline → updates Google Sheets → sends confirmation on LINE → sends HTML email report to Project Engineer.

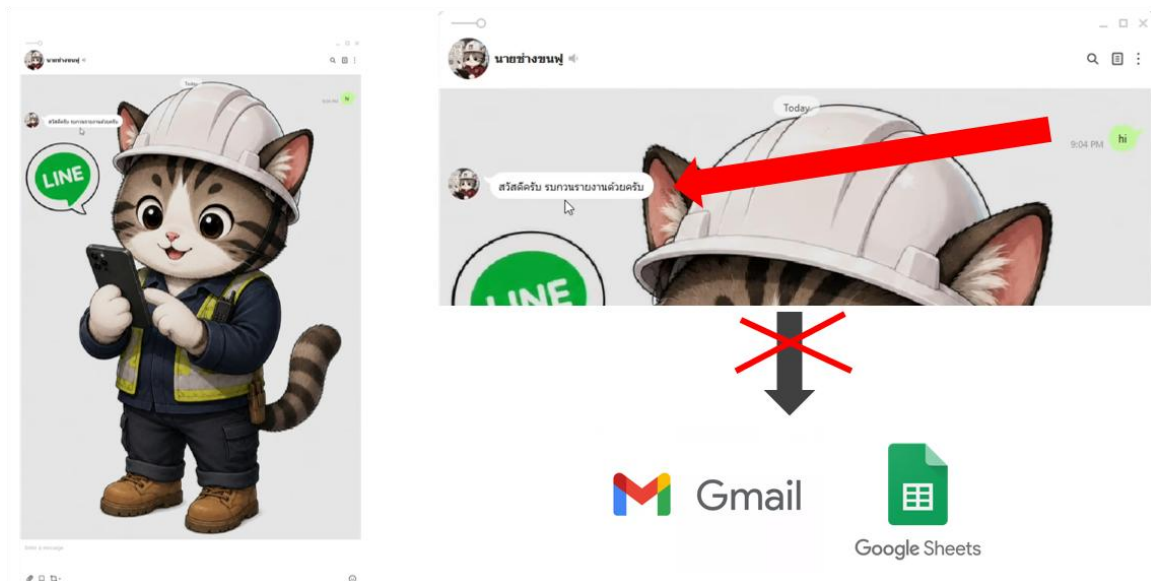


Figure 5: Scenario 2 — Successful progress update; Google Sheets is updated and Gmail report is dispatched to Project Engineer

4.3 Scenario 3: Delayed Report — Critical Path Impact

When a report indicates that a task was completed after its scheduled baseline date, the AI automatically performs a Gap Analysis to determine the delay duration and its downstream impact on dependent tasks in the Critical Path.

Example scenario:

Piling work completed on May 7, 2026 — but the CPM baseline required completion by May 5, 2026 (2-day delay). The successor task "Ground Beams & Columns" is impacted.

What the system does:

Updates Google Sheets with 100% completion status → sends a brief LINE confirmation → sends a detailed email to the Project Engineer with delay duration, impacted tasks, and recommended action plans such as authorizing Overtime (OT).

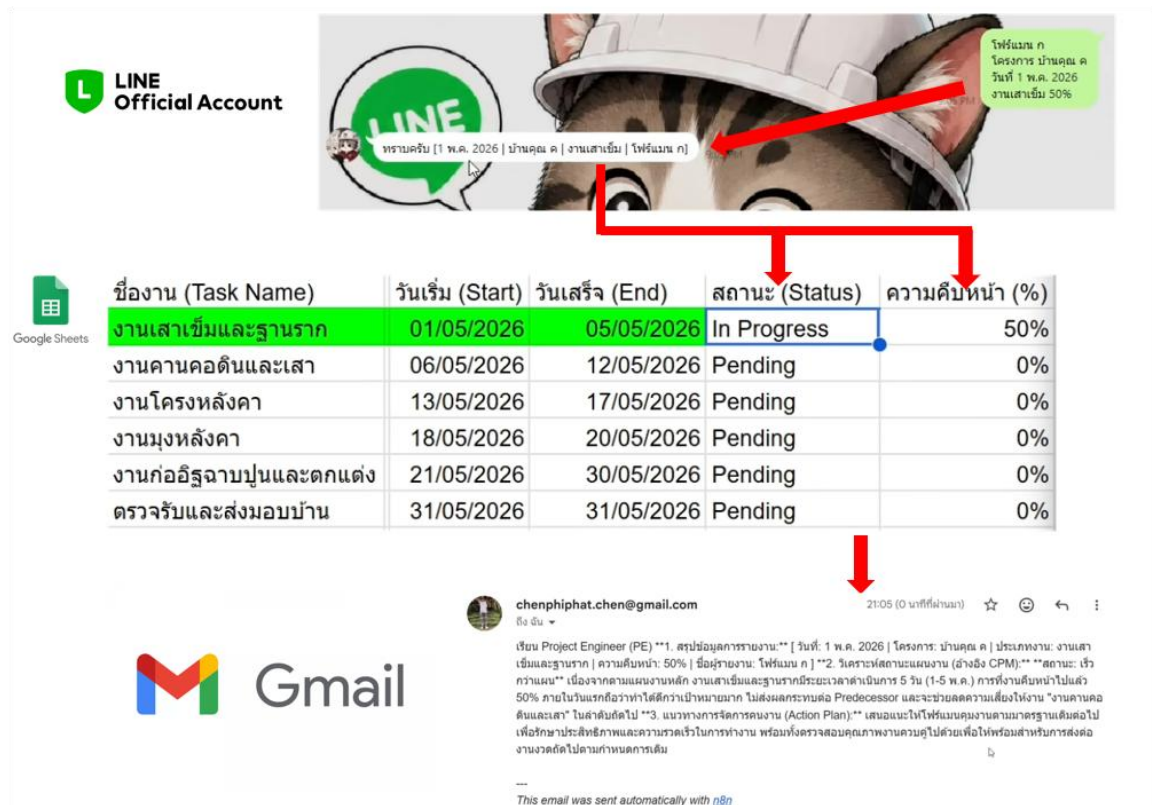


Figure 6: Scenario 3 — Delayed report: system updates database and sends Project Engineer an email with Gap Analysis and Action Plan

4.4 Scenario 4: Incomplete Report — Missing Fields

If your message is missing one or more of the three required fields (project name, date, or completion percentage), the system will NOT update the database. Instead, it will send you a LINE reply specifying exactly which information is missing and asking you to resubmit.

Example of incomplete message:

"บ้านตัวอย่าง ก งานเสร็จแล้ว" — Missing: date and completion percentage

System response:

A LINE message listing the missing fields and requesting resubmission. No data is written to Google Sheets. No email is sent to the Project Engineer.

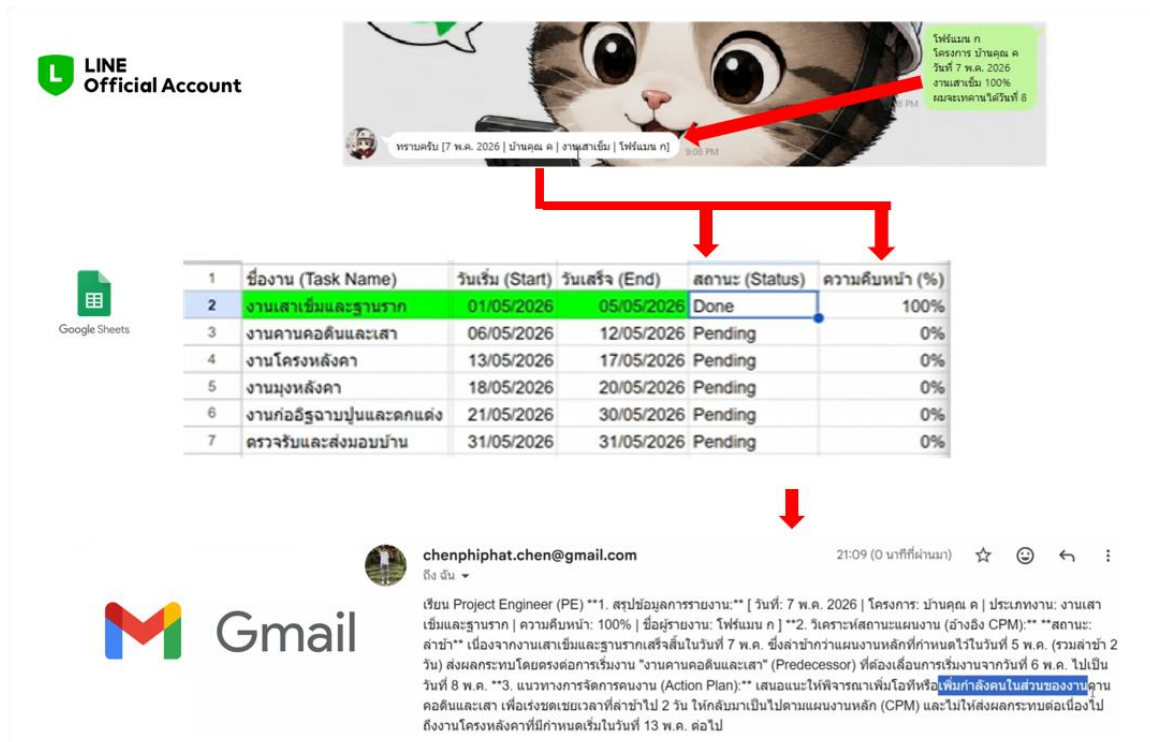


Figure 7: Scenario 4 — Incomplete data: system prompts foreman for missing fields; no backend update occurs

5. Understanding the Automated Email Report

After every successful report submission, the Project Engineer (เจ้าของงาน) receives an automated HTML email generated by the AI system. This report contains all the analytical intelligence produced by the AI Agent and is sent to the designated Gmail inbox automatically.

5.1 Email Contents

The automated email report includes the following sections:

- **Report Header:** Project name, reporting date, and foreman ID.
- **Progress Update:** Task name, reported completion percentage, and current database status.
- **Schedule Analysis:** CPM baseline comparison — whether the task is on-track, ahead, or delayed.
- **Gap Analysis (if delayed):** Exact delay duration in calendar days.
- **Critical Path Impact:** All downstream tasks affected by the delay based on predecessor-successor mapping.
- **Action Plan:** AI-generated strategic recommendations such as authorizing overtime (OT) or reallocating resources.

5.2 Sample Email Scenarios

The following screenshots show example email reports delivered to the Project Engineer for both on-schedule and delayed reporting scenarios.

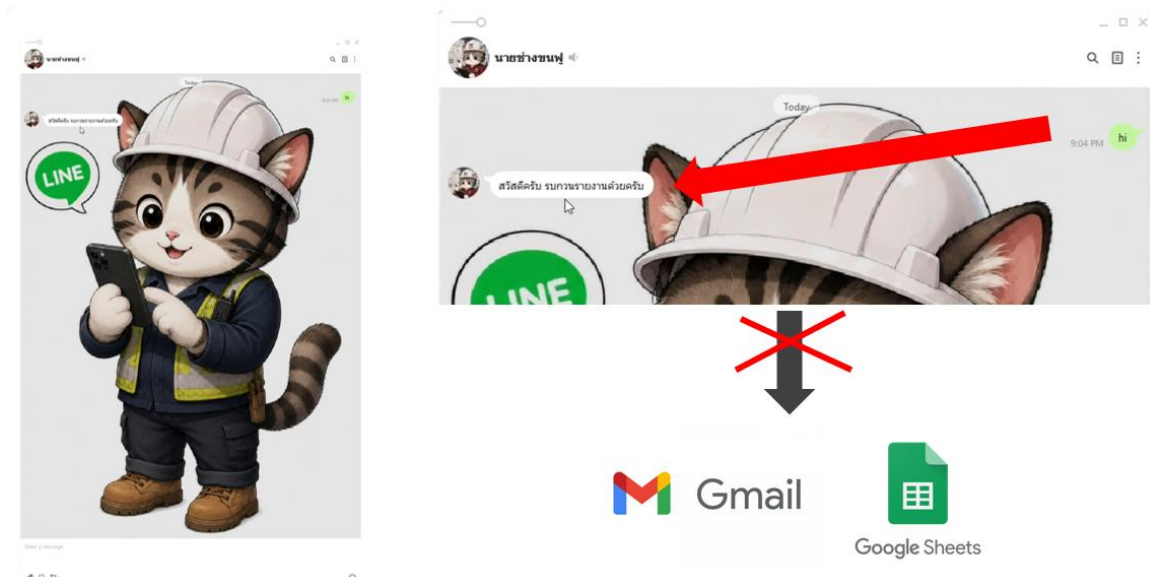


Figure 8: Email report for on-schedule update — includes progress summary and confirmation of Master Plan update

The image shows a LINE chat interface with a cat character wearing a hard hat. A red arrow points from the chat to a Gmail icon, indicating the email report. Below the Gmail icon is a Google Sheets icon.

ชื่องาน (Task Name)	วันเริ่ม (Start)	วันเสร็จ (End)	สถานะ (Status)	ความคืบหน้า (%)
งานเสาเข็มและฐานราก	01/05/2026	05/05/2026	In Progress	50%
งานคานคอดินและเสา	06/05/2026	12/05/2026	Pending	0%
งานโครงหลังคา	13/05/2026	17/05/2026	Pending	0%
งานมุงหลังคา	18/05/2026	20/05/2026	Pending	0%
งานก่ออิฐฉาบปูนและตกแต่ง	21/05/2026	30/05/2026	Pending	0%
ตรวจรับและส่งมอบบ้าน	31/05/2026	31/05/2026	Pending	0%

chenphiphat.chen@gmail.com

เขียน Project Engineer (PE) **1. สรุปข้อมูลการรายงาน** [วันที่: 1 พ.ค. 2026 | โครงการ: บ้านคุณ ค | ประเภทงาน: งานเสาเข็มและฐานราก | ความคืบหน้า: 50% | ชื่อผู้รายงาน: โฟล์แมน ก] **2. วิเคราะห์สถานะแผนงาน (อ้างอิง CPM):** **สถานะ: เร็วกว่าแผน** เนื่องจากตามแผนงานหลัก งานเสาเข็มและฐานรากมีระยะเวลาดำเนินการ 5 วัน (1-5 พ.ค.) การทำงานคืบหน้าไปแล้ว 50% ภายในวันแรกถือว่าทำได้เร็วเป็นน่ายกย่อง ไม่ส่งผลกระทบต่อ Predecessor และช่วยขจัดความเสี่ยงในงาน "งานคานคอดินและเสา" ในลำดับถัดไป **3. แนวทางการจัดการแผนงาน (Action Plan):** **เสนอแนะให้โฟล์แมนคุณงานตามมาตรฐานเดิมต่อไป เพื่อรักษาประสิทธิภาพและความรวดเร็วในการทำงาน พร้อมทั้งตรวจสอบคุณภาพงานควบคู่ไปด้วยเพื่อให้พร้อมสำหรับการส่งต่องานงวดถัดไปตามกำหนดการเดิม

Figure 9: Email report for delayed update — includes Gap Analysis (2-day delay), impacted tasks, and recommended Action Plan

6. Monitoring the Master Plan (Google Sheets)

The Master Plan is stored in a Google Sheets spreadsheet and is updated automatically every time a valid progress report is submitted. The Project Engineer can monitor the live status of all construction tasks directly in this spreadsheet at any time.

6.1 Master Plan Structure

The spreadsheet contains the following columns for each construction task:

- **Task Name** (ชื่องาน): The name of the construction task.
- **Predecessor** (งานที่ต้องเสร็จก่อน): The task that must be completed before this one can begin (CPM dependency).
- **Start Date** (วันเริ่ม): Planned start date per the CPM baseline schedule.
- **End Date** (วันสิ้นสุด): Planned completion date per the CPM baseline schedule.
- **Status** (สถานะ): Current task status — Pending, In Progress, or Done.
- **Completion %** (ความคืบหน้า): The latest reported completion percentage, updated automatically by the system.

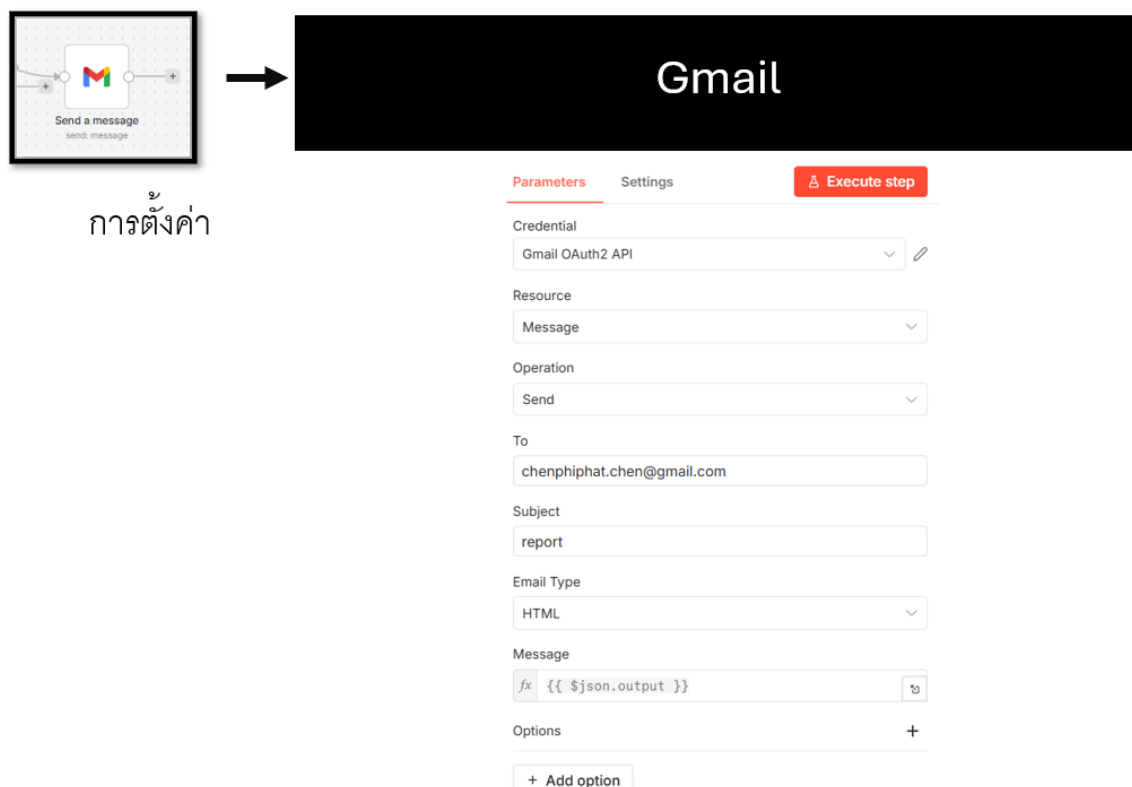


Figure 10: The CPM Master Plan table in Google Sheets showing task schedule, status, and completion percentages

7. System Workflow Overview

This section provides a technical overview of how data flows through the system from the moment a foreman sends a message to the final email delivery. This is intended for Project Engineers and Administrators who wish to understand the backend process.

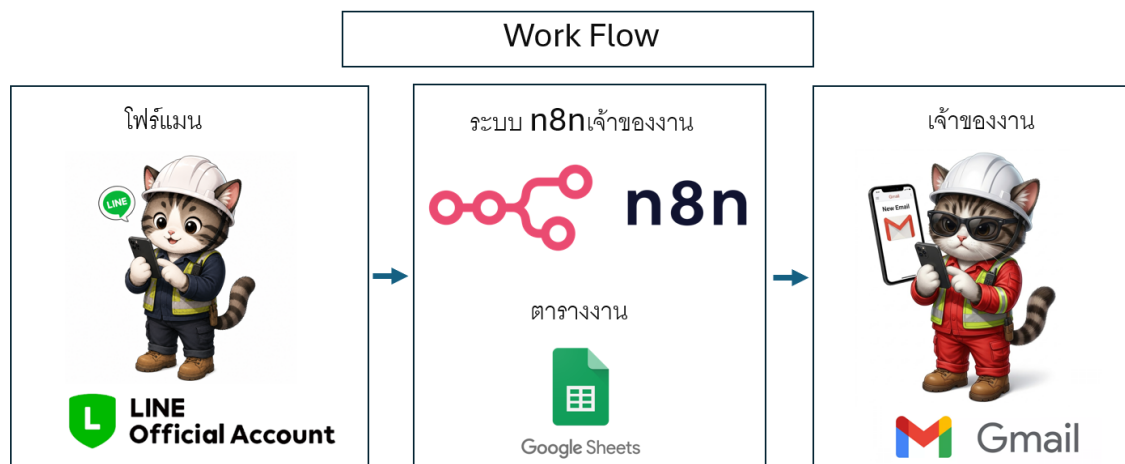


Figure 11: Complete n8n workflow canvas — all processing nodes from Webhook to Gmail

7.1 Processing Pipeline

12. **Webhook Node:** Receives the LINE message as an HTTP POST request. Extracts the message text and reply token.
13. **Edit Fields Node:** Cleans and flattens the nested JSON payload from LINE into a standardized format for AI processing.
14. **AI Agent Node (Google Gemini):** Extracts project name, date, and completion %. Compares against CPM baseline. Generates reports and action plans.
15. **IF Function Node:** Validates AI output. Routes to 'incomplete data' branch if fields are missing, or 'success' branch if complete.
16. **LINE Reply Node:** Sends confirmation (or missing-field prompt) back to the foreman on LINE.
17. **Google Sheets (check + update):** Reads current CPM schedule and commits validated progress data to the Master Plan.
18. **Gmail Node:** Sends the AI-generated HTML executive report to the Project Engineer.

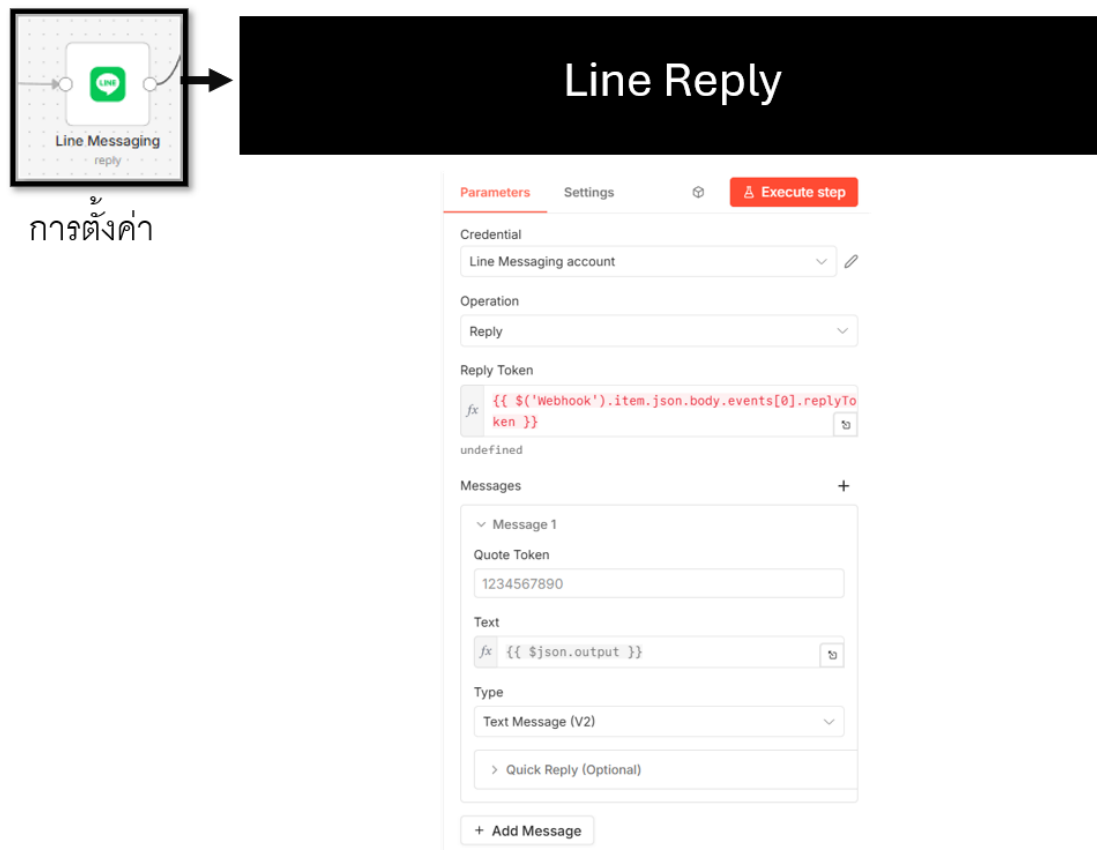


Figure 12: Gmail node configuration — automated HTML email delivery to Project Engineer

8. Troubleshooting and FAQ

Q: I sent a report but received no reply from the bot.

- Check your internet connection and ensure the LINE app is up to date.
- Verify that you have added the bot (@219gilhh) correctly as a friend.
- Try sending a greeting message ("Hi") first to confirm the bot is active.
- Contact the system administrator to check if the n8n workflow is running.

Q: The bot replied saying my information is incomplete.

- Re-read your message and ensure it contains: (1) Project Name, (2) Date, and (3) Completion Percentage.
- Resubmit your report with all three fields clearly stated.
- Example: "โครงการบ้านตัวอย่าง ก วันที่ 5 พ.ค. 2026 งานเสาเข็ม เสร็จ 80%"

Q: The Project Engineer did not receive the email report.

- Confirm that the report submission was completed successfully (you received a confirmation on LINE).
- Check the Gmail spam/junk folder for the automated email.
- Ensure the Gmail integration in n8n is still authenticated (administrator task).

Q: Google Sheets data appears incorrect or outdated.

- Only reports confirmed with a LINE acknowledgment update the sheet.
- Refresh the Google Sheets page to load the latest data.
- Contact the system administrator to verify the n8n Google Sheets credentials.

Q: I submitted the wrong completion percentage. How do I correct it?

- Submit a new report with the correct percentage for the same project and date.
- The system will overwrite the previous entry for that task.
- Alternatively, the Project Engineer can manually edit the Google Sheets entry.

9. Quick Reference Card

Use the table below as a quick reference for daily system use.

Add LINE Bot
@219gilhh or scan QR code in Figure 2
Report Format
Project Name + Date + Task + Completion %
Thai Example
"โครงการ บ้านตัวอย่าง ก วันที่ 1 พ.ค. 2026 งานเสาเข็ม เสร็จ 50%"
English Example
"Project A, 1 May 2026, Foreman A, Piling 50% complete"
Incomplete Data
Bot will reply asking for the missing fields — resubmit with all 3 fields
Email Report
Project Engineer receives automated HTML email after every valid submission
Master Plan
Google Sheets is updated automatically — no manual entry required
Delay Handling
AI generates Action Plan (e.g., Overtime) and sends to Project Engineer

10. Contact and Support

For technical issues or questions regarding the system, please contact the project team:

- **KRITTIN SUTTHIJIT** — 68070700701
- **CHENPHIPHAT SINGHARATTHANON** — 68070700703
- **PHAWITPHAT TEMUDOM** — 68070700711